Feiran Wang

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Chicago, Illinois - 60616, United States

Research Focus & Interests

- **3D Vision**: 3D Foundataion Model, Reconstruction, 3D Segmentation, Visual SLAM
- **Spatial Intelligence**: Spatial Reasoning, Vision-Language-Model, Self-Improving Agent
- **World Model**: Video Diffusion Model, Controllable Generation, Synthetic Data Generation

Education

- **Illinois Institute of Technology** — *Jan. 2024 – Current*
  Ph.D. Student in Computer Science, Advisor: Prof. Yan Yan — Chicago, United States
- **University of Illinois Urbana-Champaign** — *Aug. 2022 – Dec. 2023*
  M.Eng. in Robotics, Advisor: Prof. David Forsyth — Urbana, United States
- **Shanghai University** — *Aug. 2018 – Jul. 2022*
  B.S. in Computer Science, Advisor: Prof. Xiaoqiang Li — Shanghai, China

Publications

C=Conference, J=Journal, U=Under Review

[C.1] **Feiran Wang**, Zezhou Shang, Gaowen Liu, Yan Yan. **RayMap3R: Inference-Time RayMap for Dynamic 3D Reconstruction**. **ECCV 2026**.

[C.2] **Feiran Wang**, Junyi Wu, Dawen Cai, Yuan Hong, Yan Yan. **CogniMap3D: Cognitive 3D Mapping and Rapid Retrieval**. **ICLR 2026**.

[C.3] Junyi Wu, Van Nguyen Nguyen, Benjamin Planche, Jiachen Tao, . . . , **Feiran Wang**, Terrence Chen, Yan Yan, Ziyan Wu. **Consistent Instance Field for Dynamic Scene Understanding**. **CVPR 2026**.

[C.4] **Feiran Wang***, Jiachen Tao*, Junyi Wu*, Haoxuan Wang, Bin Duan, Zongxin Yang, Yan Yan. **X-Field: A Physically Informed Representation for 3D X-ray Reconstruction**. **NeurIPS 2025 (★Spotlight★)**.

[C.5] **Feiran Wang**, Bin Duan, Jiachen Tao, Nikhil Sharma, Gaowen Liu, Dawen Cai, Yan Yan. **ZECO: ZeroFusion Guided 3D MRI Conditional Generation**. **MVA 2025 (Oral)**.

[J.1] **Feiran Wang**, Xiaoqiang Li, Jitao Liu. **PCCN-RE: Point Cloud Colourisation Network Based on Relevance Embedding**. *IET Computer Vision*, 2022.

[U.1] Jing Gao, Wei Wang, **Feiran Wang**, Yan Yan. **LIST3R: Long-sequence Instance-aware 3D Reconstruction**.

[U.2] Jiachen Tao, Benjamin Planche, . . . , **Feiran Wang**, . . . , Terrence Chen, Yan Yan, Ziyan Wu. **From Particles to Fields: Reframing Photon Mapping with Continuous Gaussian Photon Fields**.

[U.3] **Feiran Wang**, Bin Duan, Junyi Wu, Gaowen Liu, Yan Yan. **Kepler4D: Trajectory-conditioned Future Video Generation via 4D Proxy State**.

Experience

- **Research Scientist Intern** — *May 2026 - Aug. 2026*
  Bosch Center for Artificial Intelligence, Sunnyvale, CA, Intern
  - **VLM Spatial Reasoning**: Developed vision-language models for spatial reasoning in autonomous driving, enhancing 3D scene understanding and relational grounding from multi-view camera inputs.
  - **Autonomous Driving Research**: Investigated perception and reasoning approaches for end-to-end driving, integrating VLM-based scene comprehension into the autonomous driving pipeline.
  - **Spatial Reasoning Benchmark**: Constructed a large-scale spatial reasoning benchmark, curating diverse multi-view scenarios and structured QA annotations to systematically evaluate VLM spatial understanding.
- **Robotics Software Engineer** — *May 2023 - Aug. 2023*
  Foxconn Interconnect Technology, San Jose, CA, Intern
  - **Simulation Development Platform**: Built Blender-to-MuJoCo toolchain for automatic URDF generation with joint dynamics calibration and collision geometry definition for rapid prototyping.

- ◦ **Motion Planning & Control**: Integrated MoveIt for trajectory planning and collision detection in MuJoCo; designed PD controllers with real-time state feedback for precise trajectory tracking; developed visualization dashboard for monitoring joint positions, velocities, and torques.
  - ◦ **Sim-to-Real Deployment**: Optimized control parameters through simulation-hardware closed-loop iteration; validated grasping tasks on low-cost 3D-printed hardware platform.

- **Algorithm Engineer** — *Apr. 2022 - Aug. 2022*
  NPLACE Startup, Shanghai, China, Intern
  - ◦ **Image-Based 3D Reconstruction**: Independently explored pure vision reconstruction as alternative to company's LiDAR solution; built end-to-end pipeline from multi-view image capture through MVS reconstruction to point cloud completion, providing lightweight solution for mobile 3D scanning.
  - ◦ **MVS Network Optimization**: Applied network pruning and multi-scale feature aggregation to MVSNet; compressed model via knowledge distillation achieving 10% inference speedup.

- **Backend Engineer** — *Sep. 2021 - Dec. 2021*
  ZIPLUNCH, Toronto, Canada, Intern
  - ◦ **Backend System Development**: Built restaurant order management platform backend; led database schema design and Flask RESTful API development for order lifecycle management and user authentication.
  - ◦ **Data Migration & Testing**: Designed and implemented database merge solution for legacy system migration with transaction control and validation mechanisms; built comprehensive unit test suites ensuring system stability.

Projects

- **3D Medical Image Analysis** — *Sep. 2024 - Jan. 2026*
  University of Michigan Ann Arbor, Prof. Dawen Cai — Project Page
  - ◦ **3D Medical Data Generation**: Designed ControlNet-conditioned diffusion model for controllable 3D volume generation, achieving **50**× synthetic data expansion to address medical image annotation scarcity.
  - ◦ **Semi-Automatic Annotation Platform**: Built interactive 3D annotation tool based on nnInteractive and nnUNet with morphological processing and connected component analysis, improving annotation efficiency.
  - ◦ **Neuron Instance Segmentation**: Benchmarked nnUNet and Mask2Former on neuron data; adapted architectures and loss functions for elongated topology and dense interweaving characteristics.
  - ◦ **Large-Scale Distributed Training**: Deployed 240-GPU multi-node training environment on HPC cluster with MPI optimization for large-batch parallel processing.

- **Depth Fusion Visual SLAM** — *Feb. 2023 - May 2023*
  UIUC, *Advised by Prof. Shenlong Wang*
  - ◦ **Outdoor NeRF-SLAM**: Extended NICE-SLAM to outdoor scenes with learned monocular depth prediction providing geometric priors for NeRF implicit mapping on a single GPU.
  - ◦ **Pose Optimization & Validation**: Adopted quaternion-based pose estimation with sliding window for long-sequence tracking; validated on driving data with ZED 2 stereo camera.

- **Autonomous Vehicle Perception System** — *Sep. 2022 - Dec. 2022*
  UIUC, *Advised by Prof. David Forsyth* — Project Video
  - ◦ **End-to-End Autonomous Driving System**: Built complete ROS-based autonomous driving system on real vehicle platform covering perception, planning, and control.
  - ◦ **Multimodal Perception & Localization**: Integrated YOLO detection for obstacle recognition and emergency braking; combined learning with Kalman filtering for lane tracking; implemented LiDAR SLAM.

Skills

- **Programming Languages:** Python, C++, C#, MATLAB, Javascript
- **Deep Learning & Training:** PyTorch, CUDA, TensorFlow, HPC Distributed Training
- **3D Vision & Reconstruction:** Open3D, OpenCV, SLAM, NeRF, 3DGS, VR

Academic Services & Activities

Conference Reviewer: NeurIPS 26, ECCV26, CVPR 2026, ICLR 2026, ICMR 2026, ICCV 2025, MVA 2025

Journal Reviewer: Computer Vision and Image Understanding (CVIU)

Invited Talk: Midwest Machine Learning Symposium (MMLS), 2025: Medical Imaging 3D Reconstruction

Honors and Awards

Cyrus Tang Scholarship, Illinois Institute of Technology — 2024 – 2025
International Exchange Scholarship, Shanghai University — 2020 – 2021
Academic Excellence Scholarship, Shanghai University — 2019 – 2020